

# Anmerkung Conditions Report

Rule engine covering Dachser, K+N, DHL Express, Wackler · source: assets/anmerkung.js · generated 2026-10-02

## How a row is classified

Each worksheet row is read column-wise into the four *process\** functions. A row is only classifiable when its **Stat\_Freigabe** cell equals **10** (released); otherwise the row is skipped (Wackler still runs its Kontierung check). Each forwarder compares its forwarder-specific **Differenz** columns (FR, MT, SNK, EXP, TZ, ...) against a **firing threshold**:

`hasErr(v, t) = |v| > t`

A delta that clears the threshold is called a "trigger" and contributes a label. All labels a row gathers are joined with " // " into the **Anmerkung** cell (deduplicated by join). Some rules *return early* or block later rules, because a single finding (e.g. a duplicate Fremddnummer booking) makes every other delta on the row spurious.

## Thresholds

| Constant                       | Value                                                                  |
|--------------------------------|------------------------------------------------------------------------|
| Firing threshold — Dachser     | diff  > <b>?</b> (default 0.08)                                        |
| Firing threshold — K+N         | diff  > <b>?</b> (default 0.09)                                        |
| Firing threshold — DHL Express | diff  > <b>?</b> (default 0.04)                                        |
| Firing threshold — Wackler     | diff  > <b>?</b> (default 0.09)                                        |
| Persisted per-user             | localStorage key <code>&lt;font face='Courier' size='8'&gt;anme</code> |
| Kontierung? checks             | disabled globally — KONTIERUNG_ENABLED = fal                           |

## 1. Dachser — processDachser

Gate: Stat = 10. Reads C502/C503 (Einlagern/Auslagern), SNK, ZZ, SAM, DGR, EXP, MT, TZ, C38L, FR with weight/tier logic on the DACHSER\_BP rate-card tiers (50 kg steps → 10 000 kg, open 999 999 bucket). Cascade order matters — the first rule that fires colours the rest.

### 1.0 Blank-Tarif accounting rows (early return)

| Condition                                                                                                        | Label written to Anmerkung                                                                                                                                                                                                       |
|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TARIF cell empty AND  FR  > T; no SACH, no SERV_ART, no other diff<br>(MT/TZ/EXP/SNK/ZZ/DGR/LG/AV/SAM/SBFU/C38L) | VORHOLUNG                                                                                                                                                                                                                        |
| same empty-TARIF but SACH + SERV_ART are set                                                                     | Fremdnummer <row's ReferenzNr> bereits berechnet in RE00123xxx, ok? (falls back to the 5034xxx placeholder when ReferenzNr is empty; the Beleg it was charged in stays a placeholder, it names a document outside this workbook) |
| empty TARIF + FR trigger + departure country non-DE                                                              | kein Tarif für <Land>                                                                                                                                                                                                            |

### 1.1 Direct-cost labels (each fires independently)

| Condition                                                 | Label written to Anmerkung               |
|-----------------------------------------------------------|------------------------------------------|
| C502 Kosten DL has a non-zero value                       | Einlagern                                |
| C503 Kosten DL has a non-zero value                       | Auslagern                                |
| LG Differenz over threshold                               | Lagergeld                                |
| AV Differenz over threshold                               | Gebühr für vergeblichen Abholversuch     |
| ZZ Differenz over threshold                               | 2. Zustellung                            |
| SAM Differenz over threshold                              | Samstagszustellung                       |
| DGR Differenz over threshold (skipped when TARIF is zero) | Gefahrgut-Zuschlag                       |
| SBFU Differenz over threshold                             | SBFU-Bescheinigung f. Umsatzsteuerzwecke |
| C38L Differenz over threshold                             | Maut von/bis NL                          |

### 1.2 SNK — switch on billed SNK\_DL

| Condition                                     | Label written to Anmerkung                                                                                       |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| SNK_DL = 190 or 95                            | AUSFALLFRACHT                                                                                                    |
| SNK_DL = 130                                  | Standgeld                                                                                                        |
| SNK_DL = 75 + SNK_TARIF > 0 + SNK diff over T | Differenz Hebeebuehnen-Zuschlag                                                                                  |
| SNK_DL = 75 + SERV_ART = K1AV                 | Speditionskosten gem. Text                                                                                       |
| SNK_DL = 75 (no tariff backing)               | Ausfallfracht/Schadensersatz                                                                                     |
| SNK_DL = 11 (+ diff over T)                   | Differenz Telefonische Zustellankündigung - Laderaumzuschlag                                                     |
| SNK_DL = 14 (+ diff over T, TARIF not zero)   | Differenz Automatische Zustellterminvereinbarung - Laderaumzuschlag (K1AV → Differenz Laderaumkostenentwicklung) |
| SNK_DL = 5 + SERV_ART = K1AV (SNK_TARIF=0)    | Admin Zeitfensterbuchung Handel                                                                                  |
| SNK_DL = 5 + SERV_ART = K1AV (SNK_TARIF>0)    | Differenz Admin Zeitfensterbuchung Handel - Laderaumzuschlag                                                     |

|                                                                  |                                                                     |
|------------------------------------------------------------------|---------------------------------------------------------------------|
| SNK_DL = 5 (+ diff over T); skipped when TARIF zero              | Differenz Automatische Zustellterminvereinbarung - Laderaumzuschlag |
| SNK_DL = 9 (+ diff over T, TARIF not zero)                       | Differenz Telefonische Zustellterminvereinbarung - Laderaumzuschlag |
| other SNK_DL + SERV_ART = K1AV, or non-integer SNK_DL & SNK_DIFF | Differenz Laderaumkostenentwicklung                                 |
| other SNK_DL (+ diff over T)                                     | Differenz Automatische Zustellterminvereinbarung - Laderaumzuschlag |
| Non-integer SNK_DL and SNK_DIFF rounds to 5/9/11/14 (±0.05)      | code re-derived from SNK_DIFF, then the switch above                |

### 1.3 EXP + others

| Condition                              | Label written to Anmerkung                                        |
|----------------------------------------|-------------------------------------------------------------------|
| EXP diff over T and EXP_DL = 95        | Terminzuschlag                                                    |
| EXP diff over T (otherwise)            | Produktzuschlag                                                   |
| Empf.-Ort = LONDON, PLZ leading letter | Zone korrekt berechnet? (+ Einfuhrzollabfertigung when EXP clean) |

### 1.4 FR — weight/tier decision tree (after SNK)

| Condition                                                                        | Label written to Anmerkung                                  |
|----------------------------------------------------------------------------------|-------------------------------------------------------------|
| Anz. Sdg > 1 (multi-consignment)                                                 | hätte gebündelt werden können?                              |
| REFERENZ3 = ZW (deviating intermediate consignee)                                | Differenz aufgrund abweichender Zwischenempfänger <PLZ Ort> |
| SERV_ART = K1AS                                                                  | Sonderfahrt                                                 |
| SERV_ART = K1AU and  FR – 90  ≤ 0.1                                              | "Sonderfahrt" 90 EUR doppelt berechnet?                     |
| weights equal or same tier; FR > 1 (or ≥ 0.05 with no other label); ZZ diff = 35 | Fracht Differenz (ZZ=35)                                    |
| weights equal or same tier; FR > 1 (or ≥ 0.05 with no other label); otherwise    | Frachtdifferenz                                             |
| weights equal or same tier; FR < –1.0                                            | Differenz aufgrund abweichender Gewichte                    |
| both weights known, different tier, FR 0–1                                       | Frachtdifferenz                                             |
| both weights known, different tier (FR ≥ 1 or negative)                          | Differenz aufgrund abweichender Gewichte                    |
| no weight basis, FR negative (or < 1.0)                                          | Differenz Frachtzu/ abschlag                                |
| no weight basis, FR positive                                                     | Differenz aufgrund von abweichendem Gewicht                 |
| Condition                                                                        | Label written to Anmerkung                                  |
| SNK_DL = 14 + SNK diff over T (footer)                                           | Abholterminvereinbarung                                     |
| row has no other label, only TZ over T                                           | Differenz treibstoff                                        |

## 2. K+N — processKN

Gate: Stat = 10. Tiers from KN\_BP (like Dachser but collapses the 8500/9500 steps and opens at 99 999). FR (freight) is the lead column; EXP/MT/SNK/TZ add their own labels.

| Condition                                                                     | Label written to Anmerkung               |
|-------------------------------------------------------------------------------|------------------------------------------|
| FR over T and ReferenzNr lists multiple docs (contains ',')<br>— early return | hätte gebündelt werden müssen, ok?       |
| FR over T, TARIF blank or '-' — flat-rate freight                             | Pauschalfracht                           |
| FR over T, Amazon recipient, VKG and VKG_DL in the same weight tier           | hätte nach Amazon Tarif abrechnen müssen |
| FR over T, Amazon recipient, VKG and VKG_DL cross tiers                       | Differenz aufgrund abweichender Gewichte |
| FR > 0 and SNK_DL = 5, same tier                                              | hätte nach Amazon Tarif abrechnen dürfen |
| FR > 0 and SNK_DL = 5, cross tier                                             | Differenz aufgrund abweichender Gewichte |
| any other FR over T                                                           | Differenz aufgrund abweichender Gewichte |
| EXP over T                                                                    | FIXTERMIN                                |
| MT over T                                                                     | Mautdifferenz                            |

### SNK cascade — exact matches first, then tolerance fallback

| Condition                                             | Label written to Anmerkung         |
|-------------------------------------------------------|------------------------------------|
| SNK diff $\approx$ +9                                 | Avis, ok?                          |
| SNK diff $\approx$ -9                                 | Differenz avis                     |
| SNK diff  $\approx$ 25                                | Portalavisierung, ok?              |
| SNK diff over T and SNK_DL = 5 or 25                  | Portalavisierung, ok?              |
| SNK diff over T and SNK_DL = 9                        | Avis, ok?                          |
| SNK diff over T and SNK_DL = 12                       | B2C-zuschlag, ok?                  |
| SNK diff over T and SNK_DL = 18 (unless Avis already) | Avis, ok?                          |
| SNK diff over T and SNK_DL = 34                       | Portalavisierung, ok? // Avis, ok? |
| SNK diff over T, code not matched                     | SNK Differenz                      |
| TZ over T AND neither FR nor MT fired                 | Differenz treibstoff               |
| KOST or SACH blank/'-' (disabled)                     | Kontierung?                        |

### 3. DHL Express — processDHL

Gate: Stat = 10. Uses a **blocking chain**: the first group (EL/FR/PAL/OW/YO/YL/ND/SF + SNK) sets block, and only when nothing blocked does the AC/MT/NX/OS group fire.

| Condition                                              | Label written to Anmerkung                          |
|--------------------------------------------------------|-----------------------------------------------------|
| TARIF present but value 0 (raw contains '0' or is '-') | Fremdnummer xxx bereits berechnet in RExxx, ok?.    |
| EL Differenz over T                                    | EXTENDED LIABILITY, ok?                             |
| FR Differenz over T                                    | Differenz aufgrund von abweichendem Gewicht/Volumen |
| PAL Differenz over T                                   | nicht stapelbar ok?                                 |
| OW Differenz over T                                    | Differenz overweight, ok?                           |
| YO positive and divisible by 15                        | Non conveyable piece-weight ok?                     |
| YO over T (not a 15-multiple)                          | non conveyable piece ok?                            |
| YL over T                                              | Non-conveyable piece irregular ok?                  |
| ND over T                                              | Neutral delivery ok?                                |
| SF over T                                              | Direct signature ok?                                |
| SNK = 5                                                | DIRECT SIGNATURE , ok?                              |
| SNK = 25                                               | Limited quantities ok?                              |
| SNK = 30                                               | Elevated Risk, ok?                                  |
| SNK = 60                                               | Eelevated risk ok? // Restricted destination ok?    |
| SNK over T, code not matched                           | SNK Differenz                                       |
| below-block group — AC = 11                            | Addres Correction, ok?                              |
| below-block group — AC over T                          | Address Correction ok?                              |
| below-block group — MT over T                          | Mautdifferenz                                       |
| below-block group — NX over T                          | demand surcharge ok?                                |
| below-block group — OS over T                          | Oversize piece ok?                                  |
| no other label, TZ over T only                         | Differenz treibstof                                 |

## 4. Wackler — processWackler

Gate: Stat = 10 (otherwise only the Kontierung check runs). Fourteen numbered rules evaluated in order with several early returns. Weight-tier logic uses the Wackler rate-card breakpoints (50 kg steps → 10 000 kg, open 999 999 bucket) shared by the national and international rate sheets.

### Terminal (early-return) rules

| Condition                                                                                                | Label written to Anmerkung                      |
|----------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| TARIF '-' or '0' (non-empty), or blank + FR/MT/TZ signal — duplicate booking; all delta rules suppressed | Fremdnummer xxx bereits berechnet in RExxx, ok? |
| TARIF blank + SNK signal + no FR/MT/TZ + no known SNK/AVIS code — storage fee                            | Lagergeld, ok?                                  |
| SNK $\geq 1.0 \times \text{TARIF}$ + no FR/MT/TZ + no codes (placeholder tariff)                         | Pauschalfracht, ok?                             |

### AVIS surcharge codes

| Condition                                                | Label written to Anmerkung                    |
|----------------------------------------------------------|-----------------------------------------------|
| AVIS = 7.5 / 8.5 / 6.5 / 8.7 (sign-insensitive)          | Avis, ok?                                     |
| AVIS = -8.7 (should have used the cheap telephonic rate) | hätte Avisgebühr telefonisch abrechnen dürfen |
| AVIS = 1 (per-shipment Avisnachweis line)                | Differenz avis, ok?                           |

### SNK surcharge code book (sign-insensitive, tolerance $\pm 0.5$ )

| Condition                                             | Label written to Anmerkung      |
|-------------------------------------------------------|---------------------------------|
| SNK $\approx 38$                                      | NL-FIX                          |
| SNK $\approx 11.5$ ( $\pm 0.1$ )                      | hätte B2C-Line abrechnen dürfen |
| SNK $\approx 22$                                      | 2. Zustellung ok?               |
| SNK $\approx 25$                                      | Terminzustellung                |
| SNK $\approx 80$ on a weighed shipment (weight-gated) | Terminzustellung                |
| SNK $\approx 170$                                     | Terminzustellung                |
| SNK $\approx 180$                                     | Terminzustellung, ok?           |
| SNK $\approx 43$                                      | 2.Zustellung ok?                |
| SNK $\approx 289$                                     | Umverfügung                     |
| SNK $\approx 49$                                      | NL-12, ok?                      |
| SNK $\approx 113$                                     | NL-SPEZ, ok?                    |
| Empf. = 88499 RIEDLINGEN                              | Return, ok?                     |
| SNK $\approx 16$                                      | NIGHTFIX                        |

### Gewichte / bundling decision tree (FR + both weights real)

| Condition                                                                                                    | Label written to Anmerkung                                 |
|--------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| multi-ref, weights 1–5% apart (same or cross tier)                                                           | hätte gebündelt werden müssen                              |
| VKG and VKG_DL in different rate tiers                                                                       | Differenz aufgrund abweichender Gewichte                   |
| same tier, $\geq 6$ references                                                                               | hätte gebündelt werden müssen                              |
| pallet volume ( $\text{colli} \times 285$ ) $\geq 10\,000$ kg or $\geq 1.05 \times \text{VKG}$ , tiers apart | Differenz aufgrund abweichender Gewichte                   |
| same tier, $\geq 3$ references (fallback)                                                                    | Differenz aufgrund abweichender Gewichte                   |
| otherwise (same-tier rate-card case)                                                                         | Wackler rechnet Frachtrate für $<\text{tier}>\text{kg}$ ab |

|                                                                                 |                                                  |
|---------------------------------------------------------------------------------|--------------------------------------------------|
| cross-tier, multi-ref, VKG_DL $\leq$ 20% of VKG — TERMINAL                      | hätte gebündelt werden müssen (alone)            |
| cross-tier, multi-ref, FR > 0, weights within 20%                               | hätte gebündelt werden müssen                    |
| cross-tier, far weights or FR credit                                            | Differenz aufgrund abweichender Gewichte         |
| same-tier multi-ref with unequal whole-kg weights                               | hätte gebündelt werden müssen                    |
| same-tier multi-ref with (near-)identical weights, non-integer (volumetric) VKG | Wackler rechnet Frachtrate für <floor tier>kg ab |
| no real weights + multi-ref + FR                                                | hätte gebündelt werden müssen                    |

### Fallback / additive rules

| Condition                                                                                                                                                       | Label written to Anmerkung                        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| FR over T, no Gewichte/bundling/return fired                                                                                                                    | Frachtdifferenz                                   |
| MT over T                                                                                                                                                       | Mautdifferenz                                     |
| SNK  $\geq$ 5.0, unrecognised code, not bundled                                                                                                                 | SNK Differenz                                     |
| SNK $\approx$ -150 ( $\pm 2$ )                                                                                                                                  | hätte Hebebühne abrechnen dürfen                  |
| TZ  $\geq$ 2.0, not on same-tier 'Wackler rechnet' row, not an FR credit — on a Hebebühne credit row the fuel gate queries the surcharge instead of flagging it | Differenz Energiezuschlag // Dieselizeuschlag ok? |
| NL-FIX + FR + blank KOST/SACH — 'Zone korrekt?' prepended, Frachtdifferenz stripped                                                                             | Zone korrekt? // ...                              |
| only rule left is TZ over T                                                                                                                                     | Differenz Energiezuschlag                         |
| KOST & SACH both blank/X (disabled)                                                                                                                             | Kontierung?                                       |

### Wackler constants

| Constant                                                            | Value         |
|---------------------------------------------------------------------|---------------|
| TZ  trigger (rule 11)                                               | 2.0           |
| SNK  noise floor (rule 10)                                          | 5.0           |
| SNK $\approx$ -150 $\pm$ 2 $\rightarrow$ Hebebühne                  | 150 $\pm$ 2.0 |
| SNK $\geq$ ratio $\times$ TARIF $\rightarrow$ Pauschalfracht        | $\times$ 1.0  |
| SNK $\approx$ 80 $\pm$ 0.5 $\rightarrow$ Terminzustellung (weighed) | 80 $\pm$ 0.5  |
| Multi-ref same-tier upper bundle weight                             | 1000          |
| Cross-tier near band (bundling)                                     | 0.2           |
| VKG_DL $\leq$ fraction of VKG $\rightarrow$ terminal bundling       | 0.2           |
| Pallet volume estimate per collium (kg)                             | 285           |
| 'Near-equal' weights band                                           | 0.01          |
| Close-but-unequal bundle band                                       | 0.05          |
| Same-tier $\geq$ N references $\rightarrow$ bundling                | 6             |

## How the output is consumed

The **Anmerkung** cell is a ' // '-joined list of independent rule outputs. The diff tooling (classifyDiff) compares the engine output against the auditor's ground-truth column after normalising phrase spelling/case/whitespace (normPhrase), and buckets each row as **correct / missed / overfired / wrong**. Phrase-level decomposition then reports which specific phrase the engine missed (*missing\_phrases*) and which it over-fired (*extra\_phrases*). That comes right from the A/B training pairs — a rule change is only accepted when it re-validates against the git-ignored Soll-Ist workbooks.

Control surface: scripts/anmerkung/verify.mjs smoke-tests that the four process\* functions and window.AnmerkungV5 exist; the same periods feed the diagnostics facade. Firing thresholds and the Wackler constants above are read live out of assets/anmerkung.js at build time, so this report tracks the engine — but the cascade descriptions are maintained by hand in scripts/anmerkung/report\_conditions.py and should be kept in sync when rules change.